4. What is the problem you are solving? Please describe.

Powered roof supports (shields) in underground longwall coal mines hold up the immediate roof through hydraulic legs, and changes in leg pressure are among the earliest observable indicators of changing roof load, hydraulic faults, and support malfunction. In many operations, leg pressures are viewed on shield-control displays or reviewed after the shift, and alarm logic generally relies on fixed pressure thresholds. Such thresholds cannot adequately characterise many abnormal patterns, including gradual pressure build-up that remains below the limit, leakage-like pressure decay, unequal loading between legs, and localised loading across adjacent shields. A further difficulty is that sensor faults, stuck readings, and communication interruptions can resemble genuine pressure abnormalities, causing either missed warnings or nuisance alarms that reduce operator confidence. Commercial monitoring systems are largely imported, may be costly to adapt to Indian mine conditions, and rarely explain why an alert was generated. In addition, labelled data on abnormal shield behaviour are scarce, which makes it difficult to develop and objectively evaluate data-driven detection methods. Therefore, the proposed project aims to develop an indigenous, affordable, and explainable monitoring approach that detects and interprets abnormal shield-pressure behaviour early, distinguishes process anomalies from measurement faults, and supports safer and more productive longwall operations.

5. Explain the solution proposed.

The proposed solution, MineShield AI, is an IoT-enabled monitoring and analysis system for the hydraulic leg pressures of longwall shields. In its envisaged field form, pressure transducers on each shield leg will feed a local acquisition device and an edge gateway, which will transmit validated telemetry to a surface monitoring application. The work completed so far is a locally runnable software research prototype in which a simplified engineering model simulates a representative longwall face, including the advance, setting, and loaded cycle of each shield, roof-load variation along the face, hydraulic response, measurement noise, and injected abnormal events and sensor faults with known ground truth. All readings pass through a common telemetry schema with validation of units, ranges, duplicates, stale timestamps, and missing data, so that real sensor data can later use the same interface. The analysis follows a layered approach: engineering rules detect high pressure, rapid pressure rise or decay, leg imbalance, persistent deviation, stuck sensors, isolated spikes, and data loss; an Isolation Forest model trained only on normal-operation data provides an anomaly score; and an event-correlation layer combines evidence across legs and adjacent shields. Each alert is accompanied by an explanation derived from the actual readings, rule triggers, and model scores, together with data-quality information and recommended verification steps. A reproducible evaluation framework compares the approach with fixed-threshold and statistical baselines on held-out synthetic scenarios. The system is intended to support, not replace, qualified personnel and approved safety procedures, and field validation with certified hardware and mine data remains a future stage.

6. Describe the product/service.

The proposed product is an explainable shield-pressure monitoring platform for underground longwall mines, delivered initially as software and later integrated with certified pressure sensors and gateways.

The current prototype provides a web-based dashboard showing a schematic longwall face coloured by shield status, a face-wide pressure heatmap, leg-pressure trends, anomaly scores, telemetry device status, and an alert list. Selecting an alert opens an investigation view containing the time-series evidence, the rules that fired with observed and threshold values, model scores, feature values, neighbouring-shield behaviour, data-quality status, and recommended verification steps. Alerts are classified as process anomalies, sensor faults, or telemetry interruptions and are ranked using a transparent prioritisation score rather than an uncalibrated hazard probability. The platform also includes a scenario simulator for training and demonstration, data export, and a built-in evaluation module. Following field validation, the product is intended to be offered as a monitoring software licence with installation, calibration, and technical support services for mining companies.

7. Identify the sector to which the product/services relate.

- Underground Longwall Coal Mining
- Mine Safety and Strata Control
- Industrial IoT and Sensor Networks
- Artificial Intelligence and Data Analytics
- Mining Equipment Monitoring and Maintenance
- Simulation and Digital Twin Technologies
- Mining Software and Decision-Support Systems

8. What is the unique/innovative /competitive aspect of the product/service?

The innovative aspect of the proposed product lies in combining engineering rules, unsupervised machine learning, and spatial context into a single explainable monitoring pipeline for longwall shield pressures. Unlike conventional fixed-threshold alarms that evaluate each reading in isolation, the proposed approach considers temporal trends, differences between the legs of the same shield, and the behaviour of neighbouring shields. A sensor-first logic separates likely measurement or communication faults from genuine pressure abnormalities, so that stuck sensors, isolated spikes, and data loss are reported as data-quality issues rather than as roof-related warnings. Every alert is traceable to the observed readings, derived features, rule triggers, and model scores, which supports engineering review and user trust. The system uses a lightweight, local-first architecture that runs on ordinary computers without cloud dependence, supporting indigenous development and lower deployment cost. Its competitive advantage is supported by a reproducible evaluation framework: on 48 synthetic test events, the combined approach detected all events with a precision of about 92%, whereas a fixed-threshold baseline detected 25% of them. These results are based on simulated data and require validation with real mine measurements.

9. Please let us know the market target customers' positioning of the product.

1. **Target Customers:** Coal mining companies operating longwall faces, mine safety and strata-control departments, longwall equipment suppliers, and mining technology integrators.
2. **Primary Applications:** Shield-pressure monitoring, early warning of abnormal loading, leg-circuit fault detection, sensor health monitoring, and shift-level operational review.
3. **Market Positioning:** An affordable, indigenous, and explainable decision-support system that complements existing shield-control equipment.
4. **Competitive Advantage:** Context-aware detection, separation of sensor faults from process anomalies, transparent alert explanations, and local deployment without cloud dependence.
5. **Initial Target Market:** Indian underground coal mines operating or planning longwall faces, beginning with pilot installations on selected shields.
6. **Market Expansion:** Wider longwall operations in India and abroad, followed by monitoring of other hydraulic mining equipment.
7. **Customer Value Proposition:** Earlier identification of abnormal behaviour, fewer nuisance alarms, reduced investigation effort, and improved safety and productivity.

10. Please explain the business plan from seeding development to commercialisation.

- **Seed Development and Feasibility Assessment:** Study longwall shield hydraulics, existing monitoring practices, and user requirements, and develop a software prototype with a simulated longwall face to establish technical feasibility.
- **Research and Product Development:** Refine the pressure model, detection rules, anomaly-scoring model, and alert explanations, and extend the platform with short-horizon pressure forecasting and adapters for external sensor data.
- **Prototype Development and Validation:** Integrate certified pressure transducers and an industrial gateway in a laboratory or surface test set-up, and validate data acquisition, telemetry handling, alert latency, and detection performance under controlled test conditions.
- **Safety, Performance and Economic Assessment:** Assess detection accuracy, false-alarm rates, reliability, and operator workload, and evaluate hardware, installation, and maintenance costs to establish the commercial feasibility of the system.
- **Pilot Implementation and Partnerships:** Collaborate with coal mining companies, longwall equipment suppliers, and research institutions to install the system on selected shields of an operating face and calibrate it with real mine data.
- **Business Model and Market Development:** Offer the platform through software licences with installation, calibration, and annual support contracts. Additional revenue streams include data-analysis services, customised integration with existing shield-control systems, and training for mine personnel.

- **Regulatory Compliance and Quality Assurance:** Obtain the required approvals for underground deployment, use intrinsically safe certified hardware, follow applicable DGMS guidelines and cybersecurity practices, and establish documented testing, calibration, and software-quality procedures.
- **Commercialisation and Market Expansion:** Begin with pilot mines in India and extend to additional longwall faces after successful validation, followed by adaptation for other hydraulic mining equipment and entry into international markets.
- **Expected Outcome:** The business aims to deliver a validated, affordable, and explainable shield-pressure monitoring system, generating revenue through software licences and services while supporting safer longwall operations and reducing dependence on imported monitoring solutions.

11. Market Strategy:

- **Target Market:** Focus on underground coal mines operating longwall faces in India, particularly operators seeking improved strata monitoring and safety analytics.
- **Customer Acquisition:** Engage coal mining companies, mine safety departments, longwall equipment suppliers, and mining technology integrators through demonstrations, technical meetings, and industry events.
- **Product Positioning:** Promote MineShield AI as an affordable and explainable decision-support system that complements existing shield-control equipment, subject to field validation and applicable approvals.
- **Demonstration and Pilot Projects:** Conduct live demonstrations using the built-in simulator, followed by pilot installations on selected shields to demonstrate detection performance and reliability.
- **Competitive Pricing:** Keep costs low through indigenous development, open-source software components, local deployment, and a modular sensor configuration.
- **Safety and Productivity Value:** Highlight earlier identification of abnormal pressure behaviour, fewer nuisance alarms, reduced investigation effort, and better data for strata-control decisions, supported by pilot evaluation results.
- **Quality Assurance:** Provide documented evaluation reports, calibration procedures, data-quality indicators, and transparent alert explanations to build customer confidence.
- **Market Expansion:** Begin with pilot mines and expand to additional longwall operations in India and abroad, and to the monitoring of other hydraulic mining equipment, after successful validation.
- **Revenue Generation:** Generate revenue through software licences, installation and calibration services, annual maintenance and support contracts, and data-analysis consultancy.

12. Technology: IIoT Sensing and Data Acquisition, Telemetry Validation, Hydraulic and Roof-Load Simulation, Time-Series Feature Engineering, Rule-Based Detection, Isolation Forest Anomaly Detection, Explainable Alerting, Python, Web Dashboard, Reproducible Evaluation

13. Final outcome/deliverable of the project:

- Software research prototype with simulated longwall face (developed).
- Telemetry schema and data validation interface.
- Explainable rule-based and AI anomaly detection.
- Web dashboard for monitoring and alert investigation.
- Reproducible evaluation framework and performance reports.
- Certified pressure sensor and gateway integration (planned).
- Pilot deployment and calibration with mine data (planned).
- Implementation guidelines and commercialisation plan.

14. Who would be the beneficiary of this innovation and why?

- **Coal Mining Companies:** Earlier detection of abnormal shield behaviour, fewer nuisance alarms, and improved safety and productivity.
- **Mine Workers and Supervisors:** Better awareness of roof-support conditions and less manual inspection in hazardous areas.
- **Strata-Control and Safety Engineers:** Explainable alerts and evidence for informed decisions on support performance.
- **Maintenance Teams:** Early identification of leakage, leg imbalance, and faulty pressure sensors.
- **Longwall Equipment Suppliers:** A value-added monitoring capability for their powered-support systems.
- **Regulatory and Safety Agencies:** Better data to support safety oversight and incident investigation.
- **Researchers and Academic Institutions:** A reproducible platform and synthetic datasets for studying shield behaviour and detection methods.
- **Indian Mining Technology Sector:** Indigenous capability that reduces dependence on imported monitoring solutions.

15. What is USP for your idea/product ?

- Detects abnormal shield-pressure patterns that fixed thresholds miss, including gradual rise, leakage-like decay, and leg imbalance.
- Separates sensor and communication faults from genuine pressure abnormalities, reducing false alarms.
- Explains every alert using the actual readings, rule triggers, and model scores.
- Runs locally on ordinary computers without any cloud dependence.
- Affordable, indigenous, and adaptable to different shield configurations and mine conditions.
- Performance is measured through a transparent and reproducible evaluation framework.

16. Mentorship and support (Required):

- Technical mentorship in longwall mining, strata control, and powered-support hydraulics.
- Access to pressure transducers, data acquisition hardware, and a hydraulic test set-up.
- Access to anonymised shield-pressure data from operating longwall mines.
- Expert guidance on intrinsically safe hardware, DGMS approvals, and certification.
- Industry collaboration with coal mining companies and longwall equipment suppliers.
- Funding support for hardware integration, pilot testing, and field validation.
- Business mentorship for commercialisation, pricing, and market access.